

KIRUTIGA SRIKANDA PRABANNA BALAN

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SUMMARY

Mechanical engineer with CAD, battery electrochemistry, and DFMEA-driven product reliability experience spanning electromechanical device design, cell characterization, and cross-functional validation. Skilled in DOE, materials analysis, and full-cycle engineering from concept through reliability testing.

EXPERIENCE

Engineering Consultant | Canival Medical Inc.

Sep 2025 - Present

- Reduced product risk 70% by leading DFMEA under ISO 13485, identifying and mitigating failure modes with RPN above 300 and applying Design for Reliability principles across mechanical and materials systems.
- Authored GD&T-compliant engineering drawings for leaflet components for production release and oversaw stent mechanical drawings, applying DFM and DFA principles to ensure design intent was manufacturable.
- Investigated galvanic corrosion and material degradation mechanisms to evaluate component performance under operational conditions, informing design modification recommendations.

Electrochemistry Intern | Electra

Jun 2025 - Aug 2025

- 2x improvement in battery cell dendrite suppression by designing and executing a DOE screening 25+ electrolyte additives to optimize electrochemical deposition behavior and process stability.
- Developed quantitative defect and morphology metrics to characterize cell-level variability and track electrochemical performance across experimental conditions using JMP statistical analysis.

Graduate Research Assistant | UT Austin, Aerospace Engineering

Jan 2022 - May 2025

- Designed microfluidic device geometries in SolidWorks and fabricated full assemblies via photolithography, HF wet etching, and PDMS soft lithography in a Class 100 cleanroom, optimizing design parameters for yield and device reliability.
- 4x fabrication throughput improvement and defectivity reduction through DOE-driven optimization across the full device design and build cycle.
- Characterized and optimized plasma bonding on the March Plasma CS170IF RIE, reducing bonding exposure from 60 s to 22 +/- 2 s to eliminate device leakage.

Graduate Research Assistant | UT Austin, Materials Science and Engineering

Jan 2020 - Dec 2021

- Developed GITT-based electrochemical models to generate OCV profiles and improve battery cell lifetime prediction, providing a quantitative framework for cell performance characterization.
- Built a custom technoeconomic analysis framework in Excel to model commercial-scale manufacturing costs for nanostructured composite foil Li-ion battery anodes, calculating material, energy, labor, and capital costs per cell as a function of production volume after evaluating Argonne National Lab's BatPac tool.

EDUCATION

MS, Aerospace Engineering | University of Texas at Austin

2022-2025

MS, Materials Science and Engineering | University of Texas at Austin

2020-2021

BTech, Mechanical Engineering | Amrita University

2015-2019

- SAE SUPRA (2017): Designed steering subsystem in SolidWorks with GD&T-annotated drawings; authored part-level eBOM with materials, process, and fastener costing; conducted DFMEA across 6 subsystems reducing RPNs through design modifications and material analysis.

SKILLS

Design & Validation: SolidWorks (3D CAD), DFMEA (ISO 13485), Design for Reliability, DOE, SPC, root cause analysis, SOP development, JMP

Battery & Electrochemistry: GITT, OCV characterization, CA, Voltammetry, Gamry CC-CV protocol, custom cell build

Materials & Characterization: Galvanic corrosion analysis, optical microscopy, Dektak 6M stylus profilometry, SEM-EDS, defect analysis, plasma RIE

Simulation: COMSOL Multiphysics, MATLAB, Python (NumPy, SciPy, pandas), PDE/ODE systems

Fabrication: Class 100 cleanroom, photolithography (SU-8), HF wet etching, PDMS soft lithography, thin-film processing